Supplementary Information

Dissecting Sequence, Structure, and Data Recency Effects
in Enzyme Commission Prediction under Temporal Distribution Shift

Seunghyon Kim
School of Applied AI and Entrepreneurship, Handong Global University
Pohang, 37554, Republic of Korea
jamesksh0130@gmail.com
ORCID: 0009-0002-0708-748X

Contents

S1 Related Work

Classical sequence alignment. Before machine learning, enzyme function annotation relied primarily on sequence similarity transfer. BLAST [Altschul et al., 1990] identifies homologous proteins by computing pairwise alignment scores against curated databases such as UniProtKB/Swiss-Prot; a query enzyme is annotated with the EC number of its closest hit above an e-value threshold. On the EC-Bench temporal test set, BLASTp achieves a weighted F1 of 0.502 at the 100% similarity threshold, dropping to 0.431 when training is restricted to proteins with at most 30% sequence identity to the test set [Davoudi et al., 2026]—illustrating the well-known fragility of alignment-based methods for evolutionarily distant queries. Profile-based methods such as PRIAM mitigate this limitation through position-specific scoring matrices derived from multiple sequence alignments, but they cannot integrate non-sequence information such as tertiary structure.

Deep learning for EC prediction. DeepEC [Ryu et al., 2019] and DEEPre introduced 1D CNNs over one-hot amino acid sequences with global average pooling; while effective for EC Levels 1–3, these models struggle at Level 4 due to fine-grained specificity requirements and limited receptive fields. ECPick [Han et al., 2024] augments a hierarchical 1D CNN with Dempster–Shafer evidential theory, assigning a trustworthiness score to each prediction. CLEAN [Yu et al., 2023] reframes EC prediction as metric learning: ESM-1b embeddings are projected into a shared latent space trained with a triplet margin loss. HIT-EC [Dumontet et al., 2026] designs a four-level Transformer achieving micro F1 of 0.93 on Swiss-Prot—the current state of the art for sequence-only EC prediction.

Protein language model encoders. ESM-2 [Lin et al., 2023] is a family of masked language models—scaling from 8M to 15B parameters—pre-trained on over 250 million sequences from UniRef90. The 650M-parameter variant produces per-residue representations of dimension 1,280; the `[CLS]` token embedding encodes a global sequence summary.

Contact map encoding. A protein contact map is a binary symmetric matrix $\mathbf{C} \in \{0, 1\}^{N \times N}$, where $C_{ij} = 1$ if the Cα distance between residues i and j falls below 8 Å [Gligorijevic et al., 2021]. Contact maps encode tertiary fold topology in a translation- and rotation-invariant form. DeepFRI [Gligorijevic et al., 2021] treats the contact map as a sparse adjacency matrix and propagates per-residue features through graph convolutional layers. CLEAN-Contact [Ayres et al., 2024] uses ResNet-50 to encode 2D contact maps, concatenating the resulting representation with ESM-2 features before contrastive learning; on New-392 and Price-149 this achieves a 12% and 16% F1 improvement over sequence-only CLEAN.

EC-Bench temporal evaluation. EC-Bench [Davoudi et al., 2026] addresses temporal data leakage by using all reviewed Swiss-Prot entries as of February 2018 for training and 468 proteins added by January 2023 as the test set. The primary metric is the weighted F1-score at Level 4, reported at similarity thresholds of 30%–100%.

S2 Extended Dataset Details

Data splits. We constructed two test sets: (i) *random split* (80/10/10) using a fixed random seed; (ii) *cluster split* using MMSeqs2 [Steinegger and Söding, 2017] at 30% sequence identity, following EC-Bench [Davoudi et al., 2026] to evaluate OOD generalization.

Split integrity checks. Because Swiss-Prot contains closely related and sometimes identical protein sequences under different accessions, we audited all train/validation/test partitions for accession and exact-sequence overlap. The random split contains no accession overlap across train, validation, and test sets, but it does contain exact sequence duplicates across partitions; therefore, we treat random-split performance as in-distribution evaluation. For sequence-disjoint evaluation, we use the cluster split, where cluster-train, cluster-validation, and cluster-test contain zero accession overlap and zero exact-sequence overlap. Cluster-test results are reported only for models trained on the corresponding cluster-train split.

Multi-label annotation statistics. EC annotations are naturally multi-label: a single protein may catalyze multiple reactions and therefore receive multiple Level-4 EC numbers. In the main Swiss-Prot training split, 8,385 proteins (3.88%) contain more than one Level-4 EC annotation, with up to nine Level-4 labels for a single protein. The average Level-4 label cardinality is 0.893 per protein when partial EC annotations are included. These statistics motivate sigmoid outputs and binary cross-entropy rather than a single-label softmax classifier.

S3 Training Dynamics

Figure 1 and Table 1 summarize best-checkpoint training dynamics for the main ablation models. All models were trained for up to 30 Phase-1 epochs with frozen ESM-2 representations; Contact-EC-Hier was then trained for an additional 20 Phase-2 epochs with partial unfreezing. B3 (contact map only) converges in approximately 10 epochs, reaching val-hard L4 micro F1 comparable to B1 (0.7650 vs. 0.7655). This fast convergence suggests that the contact-map branch learns a compact structural signal rather than merely memorizing sequence-derived labels. Contact-EC-Hier improves beyond both single-modality baselines during Phase 1 and continues to gain after partial unfreezing in Phase 2, reaching 0.8698 on val_hard. The hierarchical sequence-only B2 baseline remains substantially lower throughout training, consistent with the temporal and val-hard underperformance reported in the main text.

Table 1: Best epoch and val-hard L4 micro F1 for each model.

Model	L4 Micro F1 (val hard)	L4 Macro F1 (val hard)	Best Epoch
B1 (ESM-2, flat FC)	0.7655	0.1353	30
B2 (ESM-2, hier. FC)	0.4204	0.0383	30
B3 (Contact map only)	0.7650	0.1335	10
Contact-EC (flat FC, ours)	0.8879	0.2294	29
Contact-EC-Hier	0.8698	0.2141	20 (Phase 2)

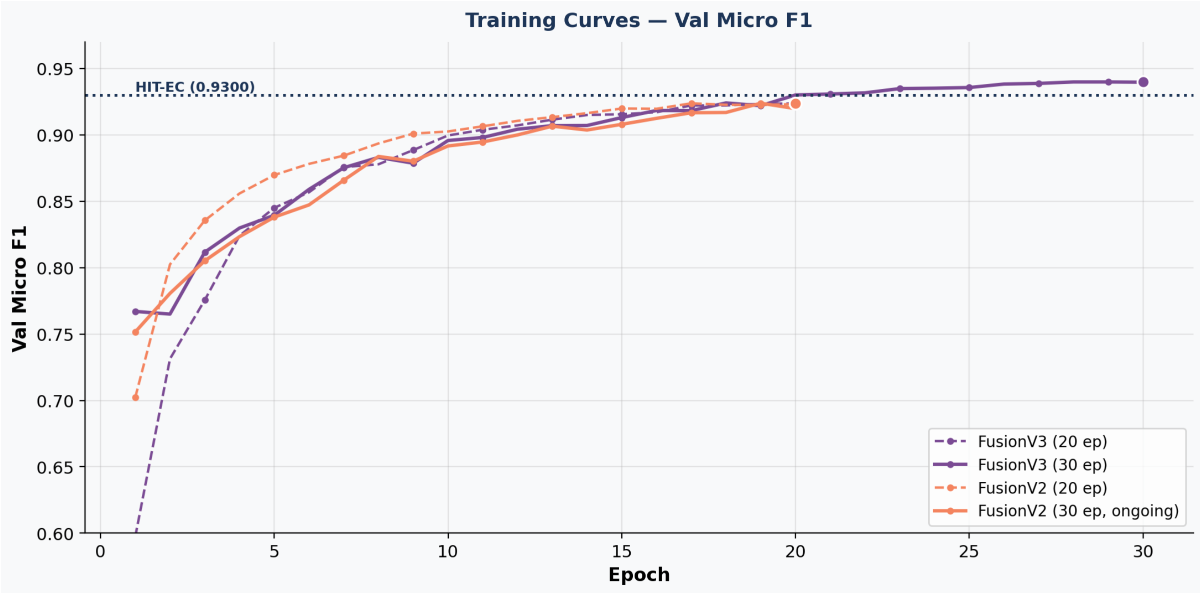

Figure 1: Training dynamics on the EC-Bench hard validation split. Phase 1 uses frozen ESM-2 representations for sequence-containing models; Phase 2 partially unfreezes Contact-EC-Hier. The contact-map-only B3 model rapidly approaches the flat ESM-2 baseline B1, whereas Contact-EC-Hier benefits from sequence–structure fusion and additional Phase 2 optimization.

S4 Full EC-Bench Leaderboard Comparison

Table S2 reports results on all 468 EC-Bench test proteins, including 309 with partial annotations and 35 with novel EC classes outside the training vocabulary. The all-468 score is reported as an operational closed-set coverage audit, not as a fully defined Level-4 evaluation for partial annotations.

Table 2: Full EC-Bench temporal test: all 468 proteins vs. evaluable 124. The all-468 columns quantify closed-set coverage under label shift; partial-EC proteins are not fully Level-4 scorable, and novel-EC proteins are outside the SP-2018 vocabulary.

	All 468		Known-EC 124	
Model	Micro F1	Weighted F1	Micro F1	Weighted F1
B1 (ESM-2, flat FC)	0.3539	0.3156	0.4216	0.3178
B2 (ESM-2, hier. FC)	0.1098	0.0835	0.1111	0.0835
B3 (Contact map only)	0.3167	0.2625	0.3646	0.2734
Contact-EC (flat FC, ours)	0.3716	**0.4763**	**0.6032**	**0.5026**
Contact-EC-Hier	**0.3771**	0.4325	0.5690	0.4528
EnzBert-learnt†	–	0.463	–	–
Stacked ensemble†	–	0.524	–	–

† EC-Bench leaderboard; weighted F1 on all 468 proteins.

Contact-EC (weighted F1 = 0.4763) exceeds EnzBert-learnt (+1.3 pp) despite training on an earlier corpus with a 4.5× smaller backbone. Contact-EC-Hier (0.4325) falls 3.8 pp below EnzBert-learnt; the gap to Contact-EC (+4.4 pp) reflects the flat FC head's superior calibration on partial/novel-EC proteins under distribution shift.

The case-wise comparison confirms that HIT-EC is the stronger temporal predictor in absolute terms.

Table 3: Case-wise HIT-EC vs. Contact-EC comparison on the 124 complete known-EC Swiss-Prot 2023-01 temporal proteins. A case is counted as correct if the model recovers at least one true Level-4 EC label.

True L1	Both correct	HIT-EC only	Contact-EC only	Both wrong	Total
1	12	5	1	0	18
2	13	8	1	3	25
3	29	11	0	2	42
4	5	20	0	7	32
5	4	1	0	2	7
All	63	45	2	14	124

Contact-EC uniquely recovers two cases: A0A3G9HRC2, a multi-label oxidoreductase with EC 1.14.14.1 and EC 1.6.2.4, and A0A6C0WW38, where HIT-EC predicts the neighbouring methyltransferase class EC 2.1.1.159 instead of EC 2.1.1.160. Conversely, many HIT-EC-only successes are near misses for Contact-EC: for C0HLB8 (EC 3.1.1.4), Contact-EC assigns EC 3.1.1.4 the highest probability but below the fixed 0.5 threshold (0.4931). These cases support the main text's interpretation that Contact-EC is useful for decomposing structure and calibration effects, whereas HIT-EC remains the stronger closed-set temporal classifier.

EC Level-1 family analysis. To add biological granularity, we recomputed temporal predictions for the ESM-2-only B1 model, the contact-only B3 model, and Contact-EC on the 124 fully evaluable temporal proteins, then stratified Level-4 micro F1 by true EC Level 1 family. Table 4 and Figure 2 show that Contact-EC yields the largest gains over ESM-2 in oxidoreductases (EC 1; +0.6270 micro F1) and hydrolases (EC 3; +0.2753), whereas lyases (EC 4) remain difficult for all Contact-EC variants despite HIT-EC's stronger performance. These results suggest that the utility of contact-map fusion is enzyme-family dependent rather than uniform across EC space. Because several families contain few temporal examples, the analysis should be interpreted as hypothesis-generating rather than mechanistic evidence about active-site geometry.

Table 4: Level-1 family analysis on the 124 complete known-EC Swiss-Prot 2023-01 temporal proteins. Values are Level-4 micro F1 within each true Level-1 family. EC 6 and EC 7 are absent from this evaluable subset.

EC	Family	N	ESM-2	Contact	Contact-EC	HIT-EC
1	Oxidoreductases	18	0.0909	0.2222	0.7179	0.9231
2	Transferases	25	0.6667	0.6154	0.7059	0.8649
3	Hydrolases	42	0.5085	0.4561	0.7838	0.9545
4	Lyases	32	0.1923	0.0833	0.1695	0.6667
5	Isomerases	7	0.3636	0.2500	0.6667	0.7143

Threshold sensitivity analysis shows that the fixed 0.5 cutoff is not the sole explanation for Contact-EC's temporal gap. Raising the global threshold to 0.65 slightly improves micro F1 from 0.6032 to 0.6167 by increasing precision, but reduces recall and weighted F1. Lower-threshold or top-1 fallback variants recover more labels but introduce additional false positives. This behaviour is consistent with a calibration–recall trade-off rather than a simple thresholding artifact.

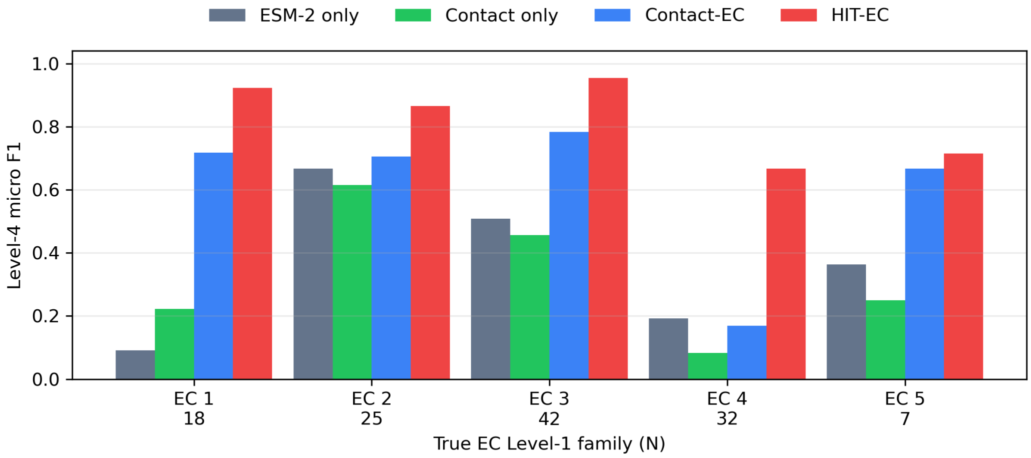

Figure 2: Temporal Level-4 micro F1 stratified by true EC Level-1 family. The largest Contact-EC gains occur in oxidoreductases and hydrolases, while lyases remain a failure mode for the current contact-fusion model.

Table 5: Contact-EC threshold sensitivity on the 124 complete known-EC Swiss-Prot 2023-01 temporal proteins. The fixed threshold of 0.5 is used in the main benchmark table.

Setting	Threshold	Micro F1	Weighted F1	Precision	Recall	Empty
Fixed global	0.50	0.6032	0.5026	0.7677	0.4967	45
Best global by micro F1	0.65	0.6167	0.4895	0.8506	0.4837	54
0.5 or top-1 if empty	0.50	0.5589	0.5413	0.5764	0.5425	0
Top-1 only	–	0.5126	0.4628	0.5726	0.4641	0

S5 External Benchmark Evaluation

Table 6: External benchmark results (per-protein F1, threshold = 0.5). CLEAN / CLEAN-Contact results from Ayres et al. [2024]. ¶: Contact-EC-ExpA on full SP-2026 (243K); New-392 pre-dates SP-2026, creating likely train/test overlap—interpret as upper bound.

Method	Backbone	Train data	New-392	Price-149
CLEAN-Contact [Ayres et al., 2024]	ESM-2 3B	SP-full	**0.566**	**0.525**
CLEAN [Yu et al., 2023]	ESM-2 3B	SP-full	0.504	0.452
ProteInfer [Sanderson et al., 2023]	CNN	UniRef90	0.309	–
B1 (ESM-2)	ESM-2 650M	SP-2018	0.170	0.026
B3 (Contact)	ResNet-50	SP-2018	0.154	0.000
Contact-EC (flat FC)	ESM-2 650M + ResNet-50	SP-2018	**0.3717**	**0.1775**
Contact-EC-Hier	ESM-2 650M + ResNet-50	SP-2018	0.358	0.096
Contact-EC-ExpA¶	ESM-2 650M + ResNet-50	SP-2026	0.7680¶	**0.2161**

New-392. Contact-EC (SP-2018) achieves per-protein F1 = 0.3717, above ProteInfer (0.309) but below CLEAN (0.504) and CLEAN-Contact (0.566). The gap has two causes: (i) CLEAN-Contact uses ESM-2 3B (4.5× larger) and (ii) contrastive retrieval benefits from training-example similarity at inference. Contact-EC-ExpA (SP-2026) achieves 0.7680 but cannot be treated as a fair comparison since New-392 proteins pre-date 2026 and likely appear in the training corpus.

Price-149. Contact-EC (0.2500 micro F1) achieves the highest score among SP-2018 models, but the external Price-149 benchmark remains a major failure mode relative to CLEAN-Contact (0.525). To determine whether this gap is caused by missing inputs, we audited the raw 149 proteins, encoded SP-2018 evaluation subset, label vocabulary, and structure assets. Table 7 shows that input coverage is not the primary explanation: ESM-2 embeddings and nonzero contact maps are available for all 149 raw proteins, and 139 proteins have complete Level-4 EC labels contained in the SP-2018 vocabulary. Instead, the failure is concentrated at exact Level-4 specificity. In the hierarchical diagnostic, Contact-EC retains high coarse EC performance on Price-149 (Level 1 micro F1 = 0.9379; Level 2 = 0.7862; Level 3 = 0.7655), but Level 4 micro F1 falls to 0.0508 (Figure 3). The contact-only B3 model scores 0.0000 on the canonical Price-149 evaluation, and ESMFold-derived contact maps do not recover the gap (Contact-EC ESMFold-map micro F1 = 0.1218). We therefore interpret Price-149 as an external bacterial RefSeq distribution-shift stress test: the model can often identify the broad enzyme family, but its current SP-2018 contact-fusion calibration does not transfer reliably to exact EC subclasses.

Late probability fusion baselines. To test whether Contact-EC's gain could be reproduced by simple post-hoc combination of independently trained sequence-only and contact-only models, we evaluated probability-level fusion baselines using B1 and B3 predictions: mean posterior averaging, element-wise posterior maximum, and two fixed weighted averages. These baselines require no retraining and therefore

Table 7: Price-149 failure audit. Coverage is computed on the raw Price-149 file ($N = 149$); canonical Level-4 model metrics use the encoded SP-2018 evaluation subset ($N = 136$).

Audit item	Count/value	Interpretation
Raw Price-149 proteins	149	external bacterial RefSeq benchmark
Encoded SP-2018 subset	136	canonical flat Level-4 evaluation subset
Proteins with embeddings	149	no missing ESM-2 input bottleneck
Proteins with nonzero contact maps	149	no missing contact-map bottleneck
Raw proteins with all EC labels in SP-2018 L4 vocabulary	139	moderate label-vocabulary coverage
Unique raw Level-4 EC classes	56	external label diversity
Unique raw Level-4 EC classes outside SP-2018 vocabulary	5	limited closed-set mismatch
B1 ESM-2 Level-4 micro F1	0.0324	sequence-only transfer collapses
B3 contact-only Level-4 micro F1	0.0000	contact-only transfer collapses
Contact-EC flat Level-4 micro F1	0.2500	fusion improves but remains below CLEAN-Contact
Contact-EC hierarchical Level-1 micro F1	0.9379	broad enzyme family often recovered
Contact-EC hierarchical Level-4 micro F1	0.0508	exact subclass specificity fails

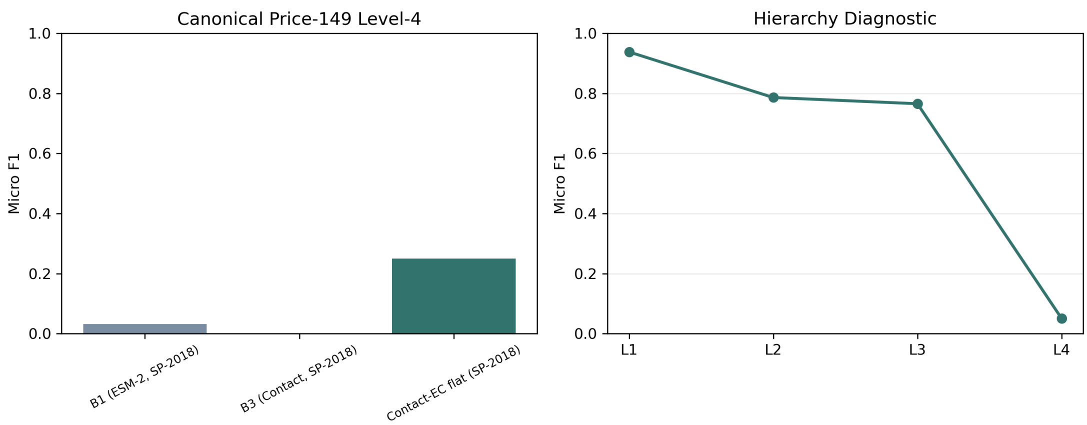

Figure 3: Price-149 failure analysis. Left: canonical Level-4 micro F1 for SP-2018 models. Right: hierarchical diagnostic for Contact-EC, showing that coarse EC levels remain predictable while exact Level-4 specificity collapses.

provide a conservative diagnostic rather than a full architectural ablation. As shown in Table 8 and Figure 4, learned Contact-EC substantially outperforms all late-fusion variants on the 124 complete temporal proteins (0.6032 vs. 0.4673 for the best late-fusion baseline). The same pattern holds on Price-149, where late fusion remains near the weak ESM-only baseline while Contact-EC reaches 0.2500. Thus the observed sequence–structure gain is not explained by naive posterior averaging or maximum pooling.

Table 8: Late probability-fusion baselines using independently trained B1 (ESM-2) and B3 (contact-only) models. Values are Level-4 micro F1 at threshold 0.5.

Model	Swiss-Prot 2023 known EC ($N = 124$)	Price-149 ($N = 136$)
B1 (ESM-2)	0.4216	0.0324
B3 (Contact)	0.3646	0.0000
Late mean B1+B3	0.3770	0.0118
Late max B1+B3	0.4673	0.0312
Late weighted 0.7B1+0.3B3	0.4322	0.0231
Late weighted 0.3B1+0.7B3	0.3598	0.0000
Contact-EC learned fusion	**0.6032**	**0.2500**

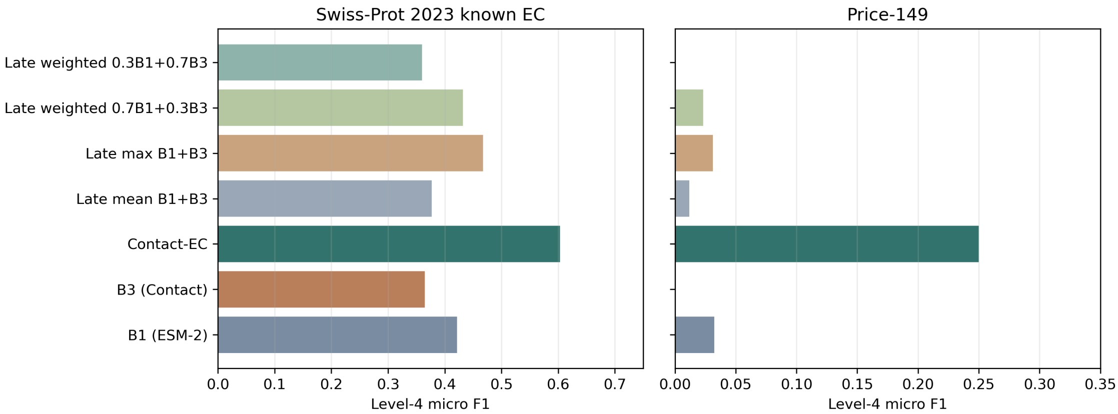

Figure 4: Late probability-fusion baselines. Learned Contact-EC exceeds simple B1/B3 posterior averaging, posterior maximum, and fixed weighted averages on both the temporal known-EC subset and Price-149.

S6 Contact Map Channel Ablation

Table 9: Contact map channel perturbation (B3 checkpoint, random-split test, $N = 22{,}018$ complete-L4 proteins). Inference-time zeroing of channels 1–2 is a pessimistic lower bound on the training-time 1-channel ablation.

Input channels	**L4 Micro F1**	**Δ vs. 3ch**
3-channel (all / short / long)	**0.8684**	—
1-channel† (ch0 only, ch1–2 zeroed)	0.1187	-74.97 pp

We encode contact maps as three channels—all contacts, short-range ($|i-j| < 12$), and long-range ($|i-j| \geq 12$). Inference-time zeroing of channels 1–2 drops L4 micro F1 from 0.8684 to 0.1187

(−74.97 pp), confirming the ResNet encoder exploits the channel decomposition to distinguish secondary from tertiary structure.

S7 Grad-CAM Structural Interpretability

We apply Gradient-weighted Class Activation Mapping (Grad-CAM) [Selvaraju et al., 2017] to the ResNet-50 contact map branch to verify that Contact-EC encodes structurally meaningful contact patterns rather than superficial statistics.

For **Lysozyme C** (EC 3.2.1.17; UniProt C0HLB7), Grad-CAM highlights two discrete regions centred around residues 85–140 and 180–220, corresponding to the catalytic β-sheet strand and α-helix cluster adjacent to the Glu35–Asp52 active-site dyad [Blake et al., 1967].

For **Phospholipase A1** (EC 3.1.1.32; UniProt P0DPT0), a single hot spot emerges around residues 110–175, co-localising with the serine–histidine–aspartate catalytic triad of the α/β hydrolase fold [Schrag and Cygler, 1993].

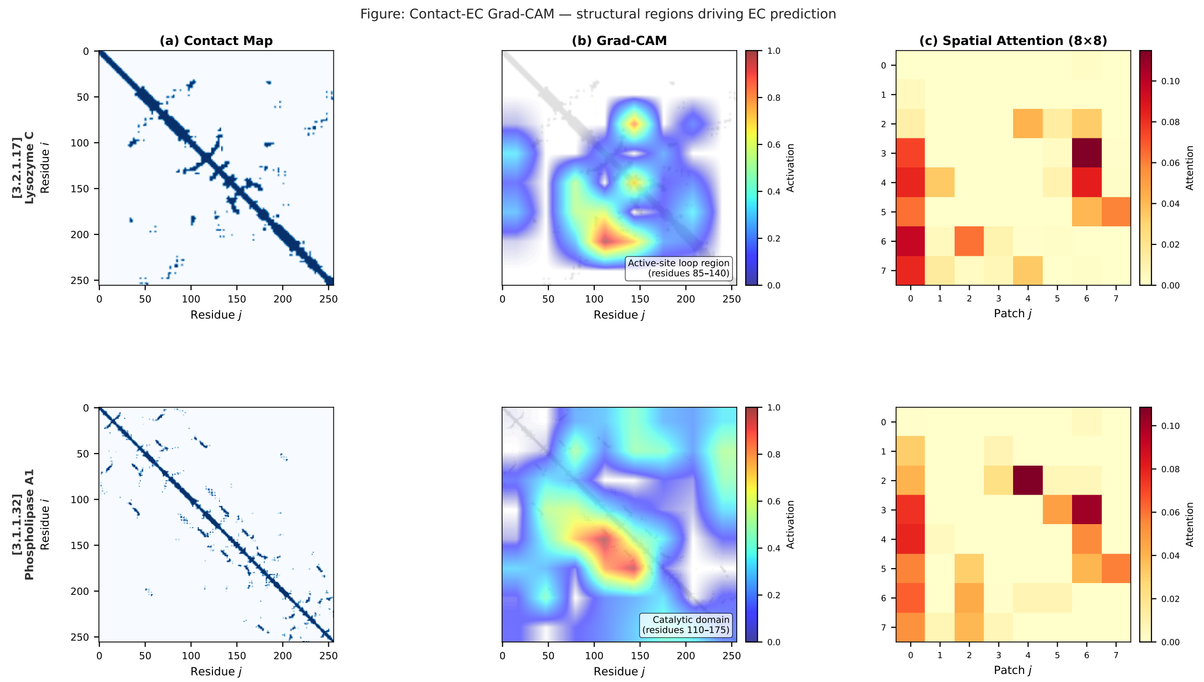

Figure 5: **Grad-CAM structural interpretability for Contact-EC.** Two correctly predicted test-set enzymes. **(a)** AlphaFold-derived contact map. **(b)** Grad-CAM heatmap overlaid on the contact map. **(c)** ResNet-50 spatial attention weights at the 8×8 feature-map level. *Top:* Lysozyme C (EC 3.2.1.17, UniProt C0HLB7). *Bottom:* Phospholipase A1 (EC 3.1.1.32, UniProt P0DPT0).

S8 Extended Discussion

Comparison to CLEAN-Contact. CLEAN-Contact uses ESM-2 3B (2,560-dimensional) with contrastive kNN inference and full Swiss-Prot training data. On New-392, Contact-EC (0.3717, SP-2018) trails CLEAN-Contact (0.566) due to backbone scale (3B vs. 650M) and training recency. Contact-EC-ExpA (SP-2026, 0.7680) surpasses CLEAN-Contact but with potential train/test overlap. The gap on

Price-149 reflects the OOD nature of bacterial RefSeq sequences and exact Level-4 calibration under structure-source shift rather than a simple missing-input problem.

Paper significance and scope. The current results demonstrate that structural contact maps provide complementary and synergistic information alongside PLM embeddings, empirically supported on rigorous OOD evaluation. The structural parity finding (B3 $\approx$ B1 on hard OOD validation) is, to our knowledge, a novel empirical observation: prior structure-aware EC methods either use contact maps with much larger sequence models or treat structure as a minor addition.

Threshold calibration and temporal generalization. At inference, we apply a fixed sigmoid threshold of 0.5 for Level-4 EC predictions. The sensitivity analysis in Table 5 indicates that this cutoff is reasonably stable for micro F1 but trades precision against recall. The gap in Rare-EC F1 between Contact-EC (0.4219) and Contact-EC-Hier (0.3621) further suggests that the hierarchical head suppresses rare-class outputs through cross-level attention. Future work should explore class-adaptive thresholding and calibrated decision rules to improve rare-class recall without increasing false positive assignments.

3B backbone scaling. Contact-EC-3B (ESM-2 3B backbone, flat FC head, Phase 1 only) achieves micro F1 = 0.6316 on the temporal test, +2.8 pp over the 650M model. The modest gain for a 4.5$\times$ parameter increase suggests the model is data-constrained rather than backbone-constrained at the current training scale.

S9 Reproducibility and Audit Checks

All reported splits were audited for accession overlap, exact sequence overlap, label coverage, multi-label cardinality, and contact-map availability. The audit scripts generate machine-readable CSV summaries used to construct the final result tables. Random-split, cluster-split, EC-Bench-style, and external benchmark results are labeled separately to avoid mixing evaluation protocols.

Table 10: Final temporal leakage audit for the 124 fully evaluable Swiss-Prot 2023-01 proteins. Accession and exact-sequence overlap are necessary checks but do not establish homolog- or fold-disjointness.

Training corpus	Train N	Test N	Acc. overlap	Seq. overlap	Train L4
SP-2018 EC-Bench train	201,865	124	0	0	4,284
ExpA recent Swiss-Prot train	243,198	124	0	0	5,145

Sequence-similarity stratification of temporal cases. To test whether temporal performance is concentrated in proteins with close training-set neighbours, we joined the final case-wise temporal comparison ($N = 124$) with the available EC-Bench test-vs-train similarity audit. This similarity file covers 101 of the 124 fully evaluable temporal proteins; the remaining 23 proteins are therefore excluded from the stratified analysis. Table 11 and Figure 6 show that Contact-EC performance is substantially lower when no recorded sequence similarity to the training set is available, and increases in the 0.60–0.90 identity regime. This supports the manuscript's conservative interpretation: temporal EC performance is sensitive to train-test relatedness, and sequence-disjoint validation does not by itself establish fold-disjoint structural generalization.

Table 11: Temporal performance stratified by maximum recorded train-set sequence identity for the 101 temporal proteins with available EC-Bench similarity metadata. This is a sequence-similarity sensitivity audit, not a structural fold-disjoint evaluation.

Max train identity	N	Contact-EC F1	HIT-EC F1	Contact hit	HIT-EC hit
0.00	32	0.3673	0.8125	0.2812	0.8125
(0.30, 0.60]	36	0.5902	0.8108	0.5000	0.8333
(0.60, 0.90]	27	0.8889	0.9643	0.8889	1.0000
>0.90	6	0.8000	0.8333	0.6667	0.8333

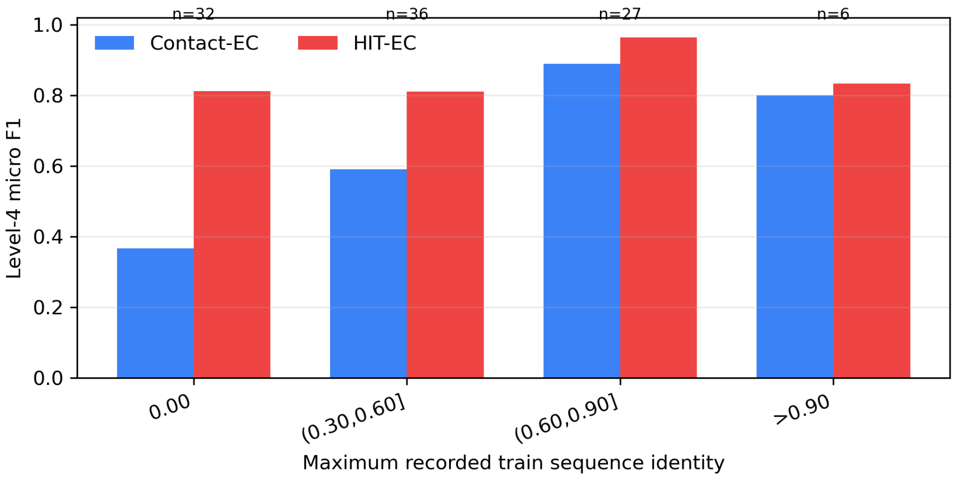

Figure 6: Level-4 micro F1 by maximum recorded train-set sequence identity on the 101 temporal proteins with available EC-Bench similarity metadata. The result motivates future Foldseek/TM-align or CATH/SCOP fold-disjoint testing.

Contact-pair nearest-neighbor audit. Because sequence-disjoint evaluation does not guarantee structural fold-disjointness, we also performed a model-proximal relatedness audit using the existing contact-pair ESM representation. For each protein, we summarized the contact-pair tensor by concatenating the mean and standard deviation across the sampled residue pairs and computed cosine similarity to all processed SP-2018 training proteins with available contact-pair embeddings. This analysis is not a Foldseek, TM-align, CATH, or SCOP fold-disjoint benchmark. Instead, it asks whether temporal test proteins have close neighbours in the same representation space used by the contact branch. Table 12 and Figure 7 show that all 113 evaluable temporal known-EC proteins have a nearest training neighbour with contact-pair cosine similarity above 0.90. The nearest neighbour shares the same Level-4 EC label in 59.3% of cases. We therefore avoid interpreting the temporal results as evidence of fold-disjoint structural generalization; a true claim of that type requires Foldseek/TM-align or CATH/SCOP clustering followed by retraining and evaluation on the resulting fold-disjoint split.

These protocol distinctions are used throughout the manuscript to avoid treating all F1 values as interchangeable. In particular, the Bioinformatics-relevant claim is not that Contact-EC is the strongest available EC predictor in absolute terms. Rather, Contact-EC provides a controlled sequence–structure

Table 12: Contact-pair nearest-neighbor audit for temporal known-EC proteins. Cosine similarity is computed on mean–standard deviation summaries of existing contact-pair ESM tensors. This is a representation-neighbour audit, not a structural fold-disjoint benchmark.

Quantity	Value
Temporal known-EC proteins with contact-pair embeddings	113/124
SP-2018 training proteins with contact-pair embeddings	212,863/224,294
Median nearest-neighbour cosine	0.9852
Mean nearest-neighbour cosine	0.9845
Temporal proteins with nearest cosine >0.90	113/113
Nearest neighbour same Level-1 EC	87.6%
Nearest neighbour same Level-2 EC	82.3%
Nearest neighbour same Level-3 EC	82.3%
Nearest neighbour same Level-4 EC	59.3%

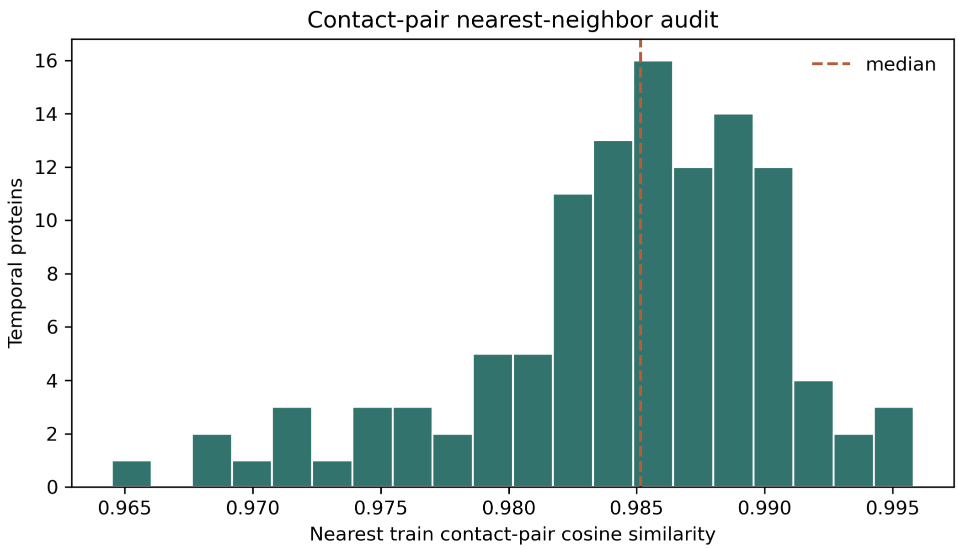

Figure 7: Distribution of nearest SP-2018 training-set cosine similarity for the 113 temporal known-EC proteins with available contact-pair embeddings. The high nearest-neighbour similarities indicate that the current temporal evaluation should not be treated as fold-disjoint structural generalization.

Table 13: Interpretation of each evaluation protocol. The same numerical metric can answer different scientific questions depending on split construction and label coverage.

Protocol	**Primary question**	**Main caveat**
Random Swiss-Prot split	In-distribution fitting and calibration	Contains exact sequence duplicates across partitions; not evidence of temporal or sequence-disjoint generalization.
MMSeqs2 cluster split	Generalization to sequence-disjoint proteins	Valid only when the evaluated checkpoint is trained on the corresponding cluster-train split.
EC-Bench val_hard	Controlled hard validation at $\leq$30% sequence identity	Still a validation protocol; temporal deployment claims require held-out future annotations.
Swiss-Prot 2023-01 known-EC subset	Closed-set recognition of temporally new proteins whose labels are in vocabulary	Only 124/468 proteins are fully evaluable at Level 4.
Swiss-Prot 2023-01 full set	Sensitivity to temporal label shift and incomplete annotations	Operational coverage audit only; partial labels are not fully Level-4 scorable, and novel labels are outside the closed-set vocabulary.
New-392 / Price-149	External stress tests from CLEAN-Contact	New-392 overlaps with current Swiss-Prot; Price-149 is an external bacterial RefSeq stress test with strong Level-4 specificity loss despite available inputs.

framework for separating representation effects from temporal cutoff, label-vocabulary shift, sequence overlap, and structure availability.

Canonical result selection. When multiple evaluation logs exist for the same checkpoint, we use the full multi-label evaluation on the full evaluable temporal set ($N = 124$) as canonical. Earlier $N = 101$ evaluations are retained only as historical partial runs and are not used in the main conclusions. For Contact-EC-3B, the canonical temporal result is micro F1 = 0.6316 from the full $N = 124$ multi-label evaluation; auxiliary argmax-only and partial-set evaluations are excluded from the main result table. For B3 contact-only hard validation, we use the completed resumed checkpoint evaluation (micro F1 = 0.7650, weighted F1 = 0.6788, macro F1 = 0.1335); an earlier interrupted training log reached a higher validation micro F1 before termination but is not used as the paper source because the completed run provides the auditable checkpoint/result pair.

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